EV – Pulse Analytics

Smart EV Charging Station Performance & Revenue Intelligence

Power BI Project Presentation

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Batch: PBI Batch-5

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Problem Statement

Electric Vehicle (EV) charging station operators generate large volumes of data from charging sessions, station locations, pricing plans, weather conditions, and calendar-based demand patterns.
However, this data is often stored across multiple sources and is not effectively utilized for decision-making.

The key business challenges are:

- ⚡ Identifying peak charging hours and high-demand locations.
- 🔋 Monitoring station utilization and charger performance.
- ⏱️ Reducing customer waiting time and congestion.
- 💰 Understanding the impact of pricing and weather on revenue and energy consumption.
- 📊 Providing management with a single, interactive view of operational performance.

Business Impact :

📈 Improve revenue visibility and pricing decisions
⏳ Reduce waiting time and charging congestion
🔌 Optimize charger and station utilization
📍 Identify high-performing and underperforming locations
📊 Enable faster data-driven management decisions

Innovation

⚡ Unified Analytics Platform – Combined charging sessions, station details, pricing, weather, and calendar data into a single Power BI model.

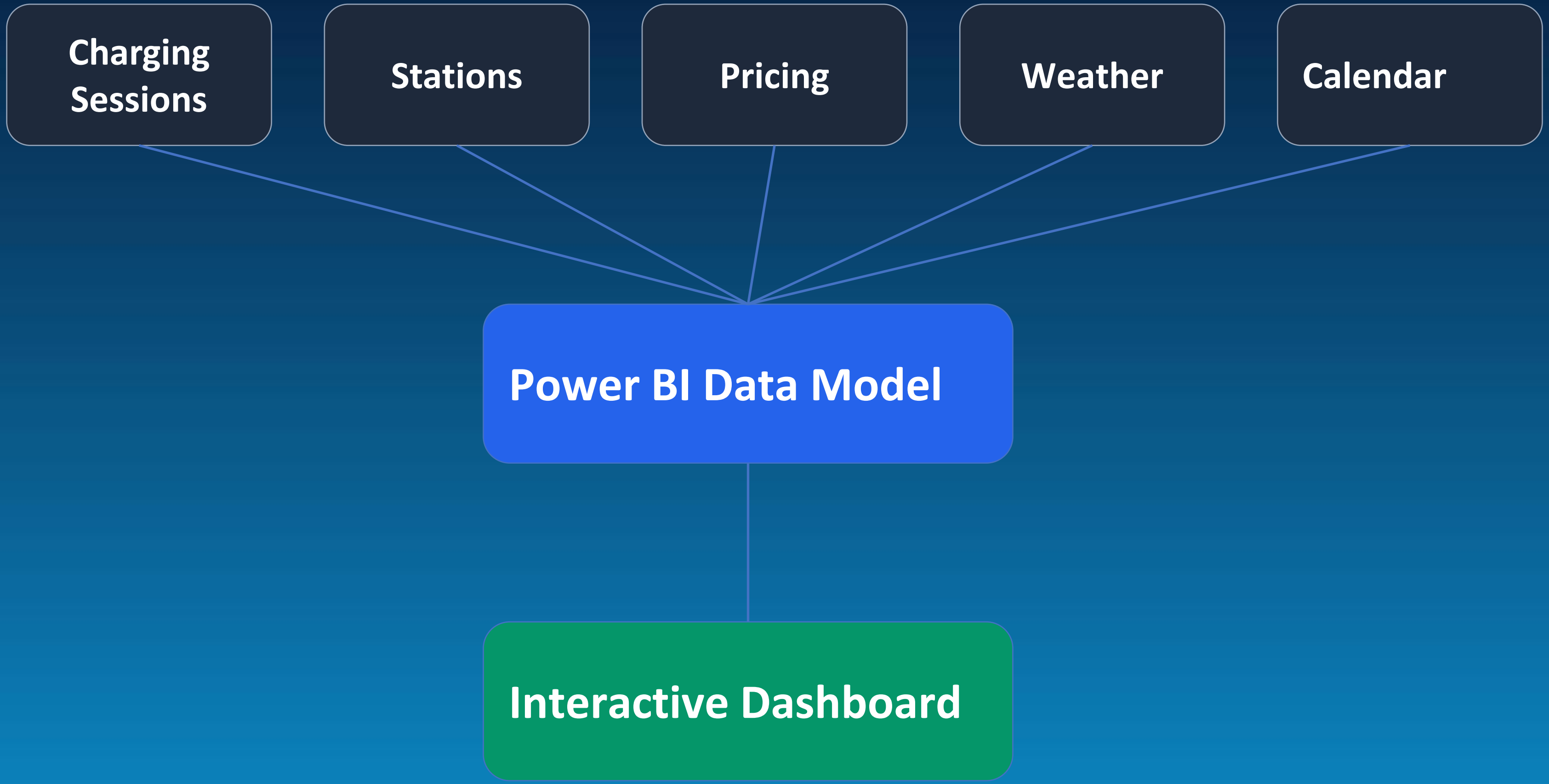
📊 Interactive Decision-Making – Designed dynamic dashboards with slicers, filters, and drill-down analysis for management-friendly exploration.

📍 Location Intelligence – Enabled area-wise and station-wise performance comparison to identify revenue hotspots and underutilized stations.

⏱️ Operational Insight – Analyzed waiting time, session completion, and charger utilization to improve customer experience and efficiency.

💡 Business-Focused Visualization – Converted raw operational data into actionable KPIs and visual insights for faster strategic decisions.

Architecture Diagram



Power BI Dashboard – KPIs

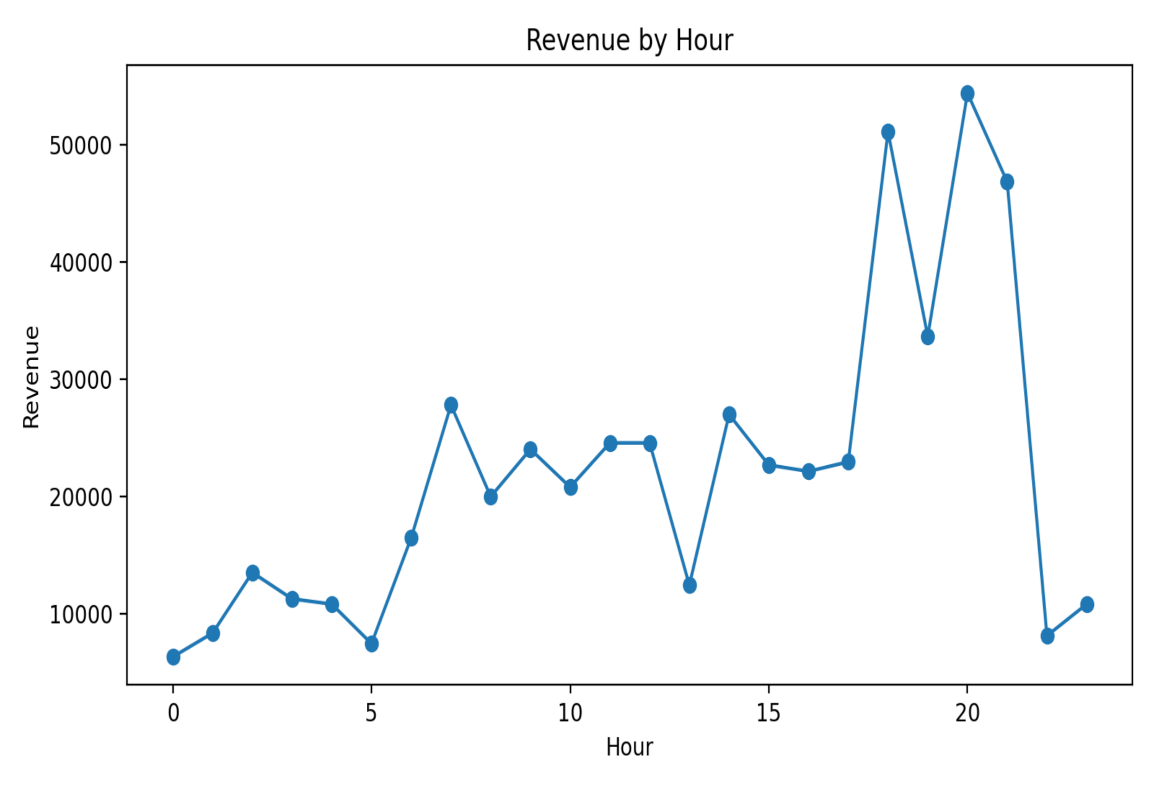
500
Total Sessions

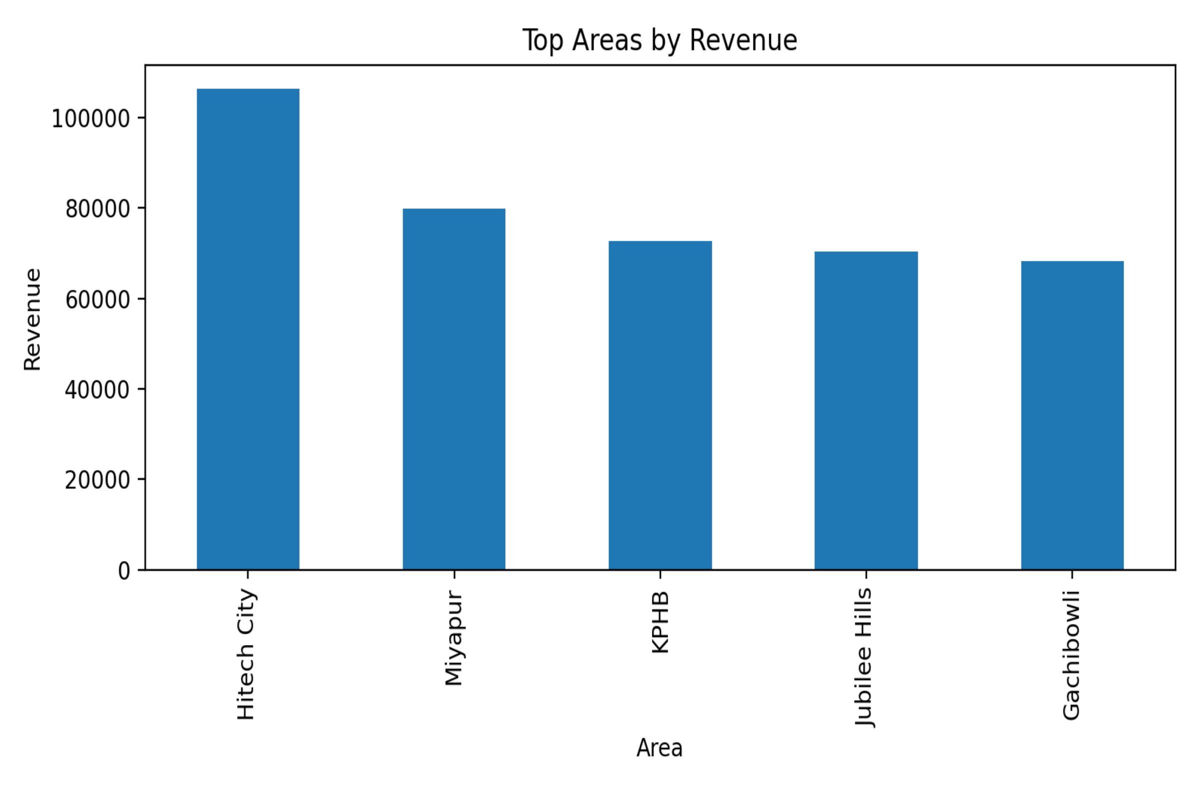
₹528,030
Total Revenue

28,305 kWh
Energy Delivered

10.6 min
Avg Wait Time

85.2%
Completion Rate





Key Insights(green)

- 📌 Peak revenue: 8 PM
- 📍 Highest revenue area: Hitech City
- 🔌 Most profitable charger : DC Fast
- 📈 Strong and consistent charging demand
- ⏱️ Congestion increases during peak hours
- ✅ High session completion rate
- 🔋 Higher utilization = higher revenue

Recommendations:

🔌 Expand DC Fast Charging

Add DC Fast chargers in Hitech City, Gachibowli, Madhapur to reduce congestion

💰 Dynamic Pricing

Offer off-peak discounts to balance demand and improve utilization

⏱️ Reduce Wait Times

Use real-time monitoring and smart queue management during peak hours

📍 Improve Operations

Optimize underperforming stations + Implement predictive maintenance

📊 Enable Real-Time Monitoring

Use Power BI dashboard for continuous tracking of revenue and utilization

Conclusion:

📊 Converted raw EV data into actionable insights
⚡ Unified operational and revenue analysis in Power BI
📈 Identified peak demand and high-performing locations
🔌 Highlighted opportunities for infrastructure optimization
⏱️ Enabled real-time KPI monitoring and faster decisions
🚀 Supports scalable and sustainable EV charging growth

💡 **EV Pulse Analytics demonstrates that data-driven monitoring and visualization can enhance operational efficiency, improve customer satisfaction, and drive strategic growth in modern EV charging networks.**